

 	Republic of the Philippines Department of Transportation
	Document No: <u>DOTr-HRDD-Forms-004</u> Rev. No.: <u>001</u> Effective Date: <u>03 August 2022</u>
	RE-ENTRY ACTION PLAN (REAP) FORM

INSTRUCTIONS: This form is used in preparing a proposed Re-entry Action Plan in applying to a Scholarship Program. Duly accomplished and signed forms will be submitted to the HRD Division along with the other documentary requirements.

Title of Scholarship Program:	Project for Human Resource Development Scholarship Program by Japanese Grant Aid ("JDS") 23rd Batch
Name of the Development Partner	Japan International Cooperation Center ("JICE") through the National Economic Development Authority ("NEDA")

I. REAP BACKGROUND AND INFORMATION	
a. REAP Title	Multi-Criteria Decision Analysis Framework for Sustainability Assessment of Prospective Port Infrastructure and Maritime Fleet Investments in the Philippines
b. REAP Objective	The study aims to develop a balanced framework for decision making by public managers and policy makers, particularly for the Philippines' Department of Transportation ("DOTr"), in systematically selecting the optimum sustainable investment option based on the priorities set forth in its mandate, its global and national commitments, and its overall policy direction. The study shall also attempt to propose sustainable thresholds to classify which options are acceptable from a government's efficiency point of view¹. The multi-criteria decision analysis framework shall be structured based on selected financing track (i.e., government-financed, locally-funded ("GF-LF"); government-financed, foreign-assisted ("GF-FA"); and public-private partnerships ("PPP")). The criteria shall factor in various economic, social, environmental, and institutional/political considerations², weighted accordingly based on an assessment to be conducted as part of the study. Subject to further analysis, a multicriteria decision-making process utilizing analytic hierarchy process ("AHP") and Technique for Order Preference by Similarity to Ideal Solution ("TOPSIS") in the context of sustainable transport is proposed. This study shall attempt to provide answers (direct or indirect) to a myriad of investment decisions including but not limited to the following questions: 1. How does a mix of economic, social, environmental, and institutional considerations factor in in the sustainability assessment of master-

	plan identified port investments which are to be implemented through (a) GF-LF; (b) GF-FA; and (c) PPP? 2. Are investments in green port infrastructure elements such as shore-side electricity-onshore power supply and installation of solar panels justifiable vis the investment costs and within the sustainability threshold? 3. Are investments in port facilities that could potentially enhance port operations such as cargo handling equipment and higher efficiently locomotives within ports better options from the government's efficiency point of view instead of investing in green port infrastructure elements such as shore-side electricity-onshore power supply and installation of solar panels? 4. Given a strong stakeholder or political resistance against a rather sustainable maritime investment, what mix of economic, social, and environmental considerations should be attained to ensure that such investment is within the acceptable sustainability threshold (i.e., what factors could justify pursuit of the sustainable investment option despite strong stakeholder or political resistance?) 5. What sustainability elements can be introduced or augmented to an existing concept to make it acceptable and within sustainability threshold? 6. Are electric-powered passenger boats and ferries better sustainable alternatives overall compared to hybrid or alternative fuel-powered fleets? Which among the available power source alternatives are within the sustainability threshold? [1] Government's efficiency point of view shall be according to government's investment priorities for undertaking (i) government-financed, locally-funded ("GF-LF") (ii) government-financed, foreign-assisted ("GF-FA"); and (iii) public-private partnership ("PPP") investments. [2] Subject to further validation and assessment to be conducted as part of the study, these factors shall include: (a) <i>Economic</i>. Upfront investment costs, operational costs, quantified economic benefits, contribution to circular economy, etc.; (b) <i>Social</i>. Comfort, accessibility, social perception and acceptability, safety and security, affordability, gender equity, etc.; (c) <i>Environmental</i>. Greenhouse gas emission reduction potential, disaster risk resilience, pollution control, waste management, etc.; (d) <i>Institutional/Political</i>. Implementation barriers, policies, directives, issuances, stakeholder resistance, availability and adaptability of technology, absorptive capacity, etc.
c. Problem/Opportunity	Policymakers are often faced with strong political and social pressure in deciding the most sustainable investment option for use in public transport systems, given that sustainable options are usually the more expensive choices and not always affordable in economic terms given the budgetary and fiscal constraints of the government. To address this, the study shall focus on the development of multi-criteria decision analysis framework for sustainability assessment of prospective Philippine port infrastructure and maritime fleet investments, with potential replicability to wider scale of project development cycle and



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across other transport sub-sectors.

The Philippines as a maritime nation comprises 2.3 million square kilometers exclusive economic zone, 36,000 kilometers of coastline, and 300,000 square kilometers of land and inland waters area. According to the Maritime Industry Authority, the Philippines as an archipelago relies heavily on a robust maritime industry. Approximately 98% of inter-island trade and 40 million local and foreign passengers are transported by sea. More than 25% of the world's 1.6 million maritime professionals are Filipino.

Based on the 2024 World Risk Report, the Philippines for the third consecutive year ranked as the most susceptible in terms of exposure and vulnerability to climate change out of 193 countries.

Based on the 2015 Philippine Greenhouse Gas Inventory Report for the Energy Sector, the Water-borne Navigation under the Transport Subsector accounts for 3,1474.98 GgCO₂-eq (10.16%), next to Road Transportation (26,723.03 GgCO₂-eq) (85.52%), and followed by Civil Aviation (1,345.26 GgCO₂-eq) (4.30%) and Railways (5.77 GgCO₂-eq) (0.02%). Energy Sector accounts for more than 50% of Philippine emissions in 2015.

Despite being a relatively minor contributor to global warming, the economic toll of climate change on the Philippines is substantial. According to the Philippines Country Climate and Development Report, by 2030, climate change is projected to impact 7.6% of the country's total GDP, amounting to PHP 1.4 trillion, and rising to 13.6% by 2040. Climate analytics suggest that the cost of inaction will stem primarily from infrastructure damage due to climate-induced disasters (up to PHP 365 billion), productivity losses from extreme heat (up to PHP 466 billion), business interruptions (up to PHP 527 billion), and relocation-related costs (up to PHP 8.9 billion) for communities displaced by climate events. These adverse effects are expected to intensify by 2050, potentially reaching 18% to 25% of GDP.

[3] Maritime Industry Authority (MARINA) of the Philippines website. (<https://marina.gov.ph/statistical-report/>)

[4] 2024 World Risk Report by the World Economic Forum. (<https://www.weforum.org/publications/global-risks-report-2024/>)

[5] Department of Energy of the Philippines. 2015 Philippine Greenhouse Inventory Report for the Energy Sector.

[6] National Adaptation Plan of the Philippines 2023-2050.

d. Description in relation to the Department i. Organizational Outcome	It is imperative for the Government of the Philippines to optimize opportunities for mitigation, facilitate transformative adaptation, and promote resilient and sustainable development. One opportunity to do this is through the proposed establishment of a multi-criteria decision analysis framework for comparing various sustainable alternatives in designing public investments and setting up acceptable thresholds for sustainable transport. Despite only second the Road Transportation in terms of emissions, a focus on maritime, particularly on port infrastructure and maritime fleet investments, pose a significant potential due to the following reasons: 1. The DOTr Planning and Project Development Office is currently undertaking upstream planning activities on (a) Maritime Connectivity and Ports Development Master Plan with the World Bank; and (b) Master Plan on Roll-on Roll-Off and Social Ports with the Asian Development Bank which aims to identify lists of port investment priorities of the Government of the Philippines, set to be completed by end of 2024 or mid-2025. Once port investments have been identified, a multi-criteria decision analysis framework is timely and suitable to help DOTr decide on the optimum sustainable options in designing these port investments. 2. The current Administration signals its commitment to modernize the maritime fleet, a.k.a. Maritime Fleet Modernization, similar to the Public Transport Modernization Program for the Road Transport Sector. Literature suggests that an all-electric fleet would be the optimal solution to meet the highest environmental standards. However, the heterogeneity of the existing fleet and fiscal constraints force policymakers to take decisions under the counterbalance of economic constraints, further suggesting the necessity of a multi-criteria decision analysis framework. 3. The compliance with the requirements of international organizations such as the International Maritime Organization where the Philippines has secured its position in the IMO Council under Category C for 2024-2025 biennium during the 33rd Session of the IMO Assembly held in December 2023 would allow Philippines to contribute to the maritime industry, not just as a top provider of top-caliber seafarers, but also at promoting safe and secure shipping operations, geared towards growth, modernization, resiliency and sustainability. 4. DOTr needs to catch-up on its maritime sector investments. As far as national budget is concerned, DOTr's highest investment to date is in the Port of Sorsogon, funded through 2019 national budget (PHP 550 Million). The share of DOTr's budget in port investments from 2010 to date has been at an average of ~5.02% (2011-2024). There has been no substantial data that could indicate how sustainability is factored in in these investments, but given the limited allocation for the sector, it is prudent to align such resources into more sustainable options.
ii. Target Competency Improvement	Through this training program, the applicant hopes to better understand the complex dynamics between and among the various economic, social, environmental, and institutional/political considerations in the development and evaluation of sustainable transport options.

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	By doing so, the applicant is hoped to be equipped with decision-making capabilities and capacities based on systematic, data-driven process, that aligns with the sustainable development priorities of the country.
e. Relation to the Government's Development Plan	The proposed Multi-Criteria Decision Analysis Framework for Sustainability Assessment is aligned with the overall strategy of the Philippine Development Plan 2023-2028, and most particularly with: Chapter 12: Expand and Upgrade Infrastructure <u>Outcome 1: Planning, programming, and asset management in infrastructure enhanced</u> - Embed resilient and innovative solutions in infrastructure design <u>Outcome 2: Seamless and inclusive connectivity via local and international linkages achieved</u> - Move people, goods, and information through modernized and expanded transport and digital infrastructure, with the active participation of the private sector - Address universal mobility and connectivity needs - Modernize the fleet and provide quality ancillary services The proposed Framework also contributes to: Chapter 15: Accelerate Climate Action and Strengthen Disaster Resilience <u>Outcome 1: Climate and disaster risk resilience of communities and institutions increased</u> - Boost multi-stakeholder partnership in building and translating knowledge to climate change adaptation and disaster risk reduction - Align environment, social, and governance (ESG) measures and investments with local adaptation and risk reduction needs and priorities <u>Outcome 3: Low carbon economy transition enabled</u> - Implement the Nationally Determined Contribution (NDC) policies and measures - Bolster private sector investments in green development - Ensure just transition of workers affected by the structural changes toward a greener, more sustainable, and low-carbon economy - Expand market opportunities for low-carbon technologies and

	products
f. Direct Beneficiary	The Department particularly the Planning and Project Development Office – Water Transport Planning Division, and the Maritime Sector, and Maritime Attached Agencies such as the Philippine Ports Authority (PPA) and Maritime Industry Authority (MARINA).
g. Gender Equality, Disability, and Social Inclusion (GEDSI)	The proposed Multi-Criteria Decision Analysis Framework for Sustainability Assessment shall entail various economic, social, environmental, and institutional/political considerations as among the prioritization criteria. In determining the appropriate weightings for such factors, a consultative process shall be undertaken, covering women and other marginalized groups (e.g., IP, PWD, LGBTQ++, etc.) and/or appropriate agencies/units, in order to validate and align the prioritization criteria with the priorities set forth in its mandate, its global and national commitments, and its overall policy direction.
h. Output	After the training program, the applicant will come up with a report detailing the concepts learned in the entirety of the program as deemed relevant to the development of the proposed Multi-Criteria Decision Analysis Framework for Sustainability Assessment of Prospective Port Infrastructure and Maritime Fleet Investments in the Philippines. A corresponding survey will be structured, for dissemination to the relevant stakeholders to serve as inputs in designing the proposed framework. It is envisioned that the proposed framework will be institutionalized upon recommendation and approval of the principals.

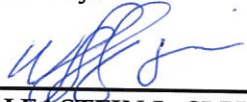
II. REAP TIMELINE AND IMPLEMENTATION

a. Measurability	- Conceptual Framework presented to identified stakeholders - Target Survey Respondents Reached - Multi-Criteria Decision Analysis Framework Tools, Templates, and Guidebook Developed - Pilot Testing of the Multi-Criteria Decision Analysis Framework Tools, Templates, and Guidebook applied to Master Plan-identified maritime project conducted - Multi-Criteria Decision Analysis Framework Tools, Templates, and Guidebook Approved - Issuance for the Adoption of Multi-Criteria Decision Analysis Framework Tools, Templates, and Guidebook Approved and Disseminated
b. Sustainability	The proposed Multi-Criteria Decision Analysis Framework for Sustainability Assessment of Prospective Port Infrastructure and Maritime Fleet Investments in the Philippines shall formally be adopted through issuance of a Department-wide Order. A reporting and monitoring mechanism shall be established as part of the report and guidebook to ensure that the framework is regularly updated, expanded to cover the whole project development cycle, and replicated across transport subsectors.

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c. Predictive Action Steps with Identified Success Indicators		
i. Phase/Stage	ii. Expected Output	iii. Timeline
Framework Development Stage (From Conceptual Design Phase, Consultation and Implementation Phase, and Approval Phase)	- Survey Design - Excel-based Tool and Templates (with Instructions and Guidebook for the Multi-Criteria Decision Analysis Framework) - Department Issuance for the Institutionalization of the Framework	9 to 12 months <i>(to be fast-tracked subject to potential technical assistance to be provided by Key Experts)</i>
d. Target Date to Start	4Q 2027	
e. Target Date to End	4Q 2028	
III. RESOURCES		
As applicable, to be sourced from General Appropriations Act – Support to Operations under Program planning and standards development for transportation and communications services, including infrastructure projects or under Maintenance and Other Operating Expenses, for expenses to be incurred in the conduct of focus group discussions with the stakeholders. Estimated Budget: PHP 70,000.00		
IV. COMMENT(S) AND RECOMMENDATION(S)		
a. Immediate Superior (Mentor)		
b. Head of the Office/Division/Service		
c. Human Resource Development Division		

Prepared by:


WALLEASTEIN L. SIGUI

Supervising Transportation Development Officer
 Office of the Undersecretary for Planning and Project Development

Recommended and Approved by:



TIMOTHY JOHN R. BATAN
Undersecretary for Planning and Project Development